HDOC Day-9

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Batch - 15

Question 1: Menu-Driven Program for Swapping Two Numbers

1. Problem Statement

Write a menu-driven C program that accepts two numbers and provides two choices:

1. Swap using a third variable.
2. Swap without using a third variable.

The program must display the values **before and after swapping**.

2. Algorithm

1. Start.
2. Input two numbers `a` and `b`.
3. Display their values before swapping.
4. Display the menu:
   - 1 → Swap using third variable
   - 2 → Swap without third variable
5. Input the user's choice.
6. If choice is 1:
   - `temp = a`
   - `a = b`
   - `b = temp`
7. If choice is 2:
   - `a = a + b`
   - `b = a - b`
   - `a = a - b`
8. Otherwise, display `"Invalid choice"`.
9. Display the values after swapping.
10. Stop.

3. Pseudocode

```
START

INPUT a, b


PRINT "Before swapping", a, b

PRINT "1. Swap using third variable"

PRINT "2. Swap without third variable"

INPUT choice

IF choice = 1 THEN

    temp = a

    a = b

    b = temp

ELSE IF choice = 2 THEN

    a = a + b

    b = a - b

    a = a - b

ELSE

PRINT "Invalid choice"

STOP

END IF

PRINT "After swapping", a, b

STOP
```

4. Flowchart

START

↓

Input a and b

↓

Display a and b
before swapping

↓

Display Menu
1. Third variable
2. Without third var.

↓

Input choice

```
└──────────┬──────────┘

     ↓

◇ Choice = 1? ◇

/            \

Yes              No

↓                ↓

temp = a         ◇ Choice = 2? ◇

a = b               /         \

b = temp          Yes          No

                   ↓            ↓

              a = a+b      Invalid Choice

              b = a-b           ↓

              a = a-b         STOP

                   ↓

┌──────────────────────────────┐

│ Display a and b │

│ after swapping  │

└──────────────┬───────────────┘

       ↓

  ┌────────────┐

  │  STOP  │

  └────────────┘
```

5. Test Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	a=10, b=20, Choice=1	Before: 10,20; After: 20,10	Before: 10,20; After: 20,10	Pass
2	a=5, b=8, Choice=2	Before: 5,8; After: 8,5	Before: 5,8; After: 8,5	Pass
3	a=-4, b=6, Choice=1	Before: -4,6; After: 6,-4	Before: -4,6; After: 6,-4	Pass
4	a=12, b=12, Choice=2	Before: 12,12; After: 12,12	Before: 12,12; After: 12,12	Pass
5	a=10, b=20, Choice=3	Invalid choice	Invalid choice	Pass

6. C Program

```
#include <stdio.h>

int main()
{
  int a, b, temp, choice;
  printf("Enter two numbers: ");
  scanf("%d %d", &a, &b);
  printf("\nBefore swapping: a = %d, b = %d\n", a, b);
  printf("\n1. Swap using third variable");
  printf("\n2. Swap without using third variable");
  printf("\nEnter your choice: ");
  scanf("%d", &choice);
  switch(choice)
    {
     case 1:
            temp = a;
            a = b;
            b = temp;
            break;
     case 2:
            a = a + b;
            b = a - b;
            a = a - b;
            break;
      default:
            printf("Invalid choice\n");
            return 0;
    }

    printf("After swapping: a = %d, b = %d\n", a, b);
    return 0;
}
```

```
  UW PICO 5.09                                                    File: 9q1.c

#include <stdio.h>

int main()
{
    int a, b, temp, choice;

    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);

    printf("\nBefore swapping: a = %d, b = %d\n", a, b);

    printf("\n1. Swap using third variable");
    printf("\n2. Swap without using third variable");
    printf("\nEnter your choice: ");
    scanf("%d", &choice);

    switch(choice)
    {
        case 1:
            temp = a;
            a = b;
            b = temp;
            break;

        case 2:
            a = a + b;
            b = a - b;
            a = a - b;
            break;

        default:
            printf("Invalid choice\n");
            return 0;
    }

    printf("After swapping: a = %d, b = %d\n", a, b);

    return 0;
}
```

7. Compilation and execution

```
[aaryanrajput@aaryans-macbook ~ % nano 9q1.c
[aaryanrajput@aaryans-macbook ~ % clang 9q1.c -o 9q1.c
[aaryanrajput@aaryans-macbook ~ % ./9q1.c
Enter two numbers: 10 20

Before swapping: a = 10, b = 20

1. Swap using third variable
2. Swap without using third variable
Enter your choice: 1
After swapping: a = 20, b = 10
aaryanrajput@aaryans-macbook ~ %
```

Compilation and execution successful

Question 2: Sum of Natural Numbers and Their Squares

1. Problem Statement

Write a menu-driven C program to calculate:

1. Sum of the first `n` natural numbers.
2. Sum of squares of the first `n` natural numbers.

Formulas

For the sum of first `n` natural numbers:

Sum = n(n + 1) / 2

For the sum of squares:

Sum of Squares = n(n + 1)(2n + 1) / 6

2. Algorithm

1. Start.
2. input `n`.
3. Display menu:
    - 1 → Sum of first `n` natural numbers.
    - 2 → Sum of squares of first `n` natural numbers.
4. Input choice.
5. If choice = 1:
    - Calculate `sum = n(n+1)/2`.
    - Display sum.
6. If choice = 2:
    - Calculate `sum = n(n+1)(2n+1)/6`.
    - Display sum.
7. Otherwise display `"Invalid choice"`.
8. Stop.

3. Pseudocode

```
START
INPUT n
PRINT "1. Sum of first n natural numbers"
PRINT "2. Sum of squares of first n natural numbers”
INPUT choice
IF choice = 1 THEN
   sum = n * (n + 1) / 2
   PRINT sum
ELSE IF choice = 2 THEN
   sum = n * (n + 1) * (2*n + 1) / 6
   PRINT sum
ELSE
   PRINT "Invalid choice"
END IF
STOP
```

4. Flowchart

```
        START
          ↓
        Input n
          ↓
   Display Menu
   1. Sum of natural nos.
   2. Sum of squares
          ↓
      Input choice
          ↓
    ◇ Choice = 1? ◇
     /          \
   Yes          No
    ↓            ↓
Sum=n(n+1)/2     ◇ Choice = 2? ◇
    ↓        /        \
 Display Sum   Yes          No
               ↓          ↓
         Sum=n(n+1)(2n+1)/6  Invalid
               ↓           Choice
          Display Sum          ↓
               ↓          ↓
            STOP
```

5. Test Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	n=5, Choice=1	Sum = 15	Sum = 15	Pass
2	n=5, Choice=2	Sum of squares = 55	Sum of squares = 55	Pass
3	n=10, Choice=1	Sum = 55	Sum = 55	Pass
4	n=10, Choice=2	Sum of squares = 385	Sum of squares = 385	Pass
5	n=3, Choice=3	Invalid choice	Invalid choice	Pass

6. C Program

```
#include <stdio.h>

int main()
{
    int n, choice;
    long long sum;

    printf("Enter the value of n: ");
    scanf("%d", &n);

    printf("\n1. Sum of first n natural numbers");
    printf("\n2. Sum of squares of first n natural numbers");
    printf("\nEnter your choice: ");
    scanf("%d", &choice);

    switch(choice)
    {
        case 1:
            sum = (long long)n * (n + 1) / 2;
            printf("Sum of first %d natural numbers = %lld\n",
                   n, sum);
            break;

        case 2:
            sum = (long long)n * (n + 1) * (2 * n + 1) / 6;
            printf("Sum of squares of first %d natural numbers = %lld\n",
                   n, sum);
            break;

        default:
            printf("Invalid choice\n");
    }

    return 0;
}
```

```
 UW PICO 5.09

#include <stdio.h>

int main()
{
    int n, choice;
    long long sum;

    printf("Enter the value of n: ");
    scanf("%d", &n);

    printf("\n1. Sum of first n natural numbers");
    printf("\n2. Sum of squares of first n natural numbers");
    printf("\nEnter your choice: ");
    scanf("%d", &choice);

    switch(choice)
    {
        case 1:
            sum = (long long)n * (n + 1) / 2;
            printf("Sum of first %d natural numbers = %lld\n",
                   n, sum);
            break;

        case 2:
            sum = (long long)n * (n + 1) * (2 * n + 1) / 6;
            printf("Sum of squares of first %d natural numbers = %lld\n",
                   n, sum);
            break;

        default:
            printf("Invalid choice\n");
    }

    return 0;
}
```

7. Compilation and Execution

```
[aaryanrajput@aaryans-macbook ~ % nano 9q2.c
[aaryanrajput@aaryans-macbook ~ % clang 9q2.c -o 9q2.c
[aaryanrajput@aaryans-macbook ~ % ./9q2.c
 Enter the value of n: 5

 1. Sum of first n natural numbers
 2. Sum of squares of first n natural numbers
 Enter your choice: 1
 Sum of first 5 natural numbers = 15
 aaryanrajput@aaryans-macbook ~ %
```

Compilation and execution successful

Question 3: Simple Interest, Compound Amount, Compound Interest and Total Payable Amount

1. Problem Statement

Write a menu-driven C program that accepts:

- Principal (`P`)
- Rate of interest (`R`)
- Time (`T`)

The menu should calculate:

1. Simple Interest
2. Compound Amount
3. Compound Interest
4. Total Payable Amount based on simple interest

Formulas

Simple Interest:

```
SI = (P × R × T) / 100
```

Compound Amount (annual compounding):

```
A = P × (1 + R/100)^T
```

Compound Interest:

```
CI = A - P
```

Total Payable Amount using Simple Interest:

```
Total = P + SI
```

2. Algorithm

1. Start.
2. Input principal `P`, rate `R`, and time `T`.
3. Display menu.
4. Input choice.
5. Calculate according to choice:
   - Choice 1 → Simple Interest.
   - Choice 2 → Compound Amount.
   - Choice 3 → Compound Interest.
   - Choice 4 → Total Payable Amount.
6. Display calculated value.

7. If the choice is invalid, display `"Invalid choice"`.
8. Stop.

3. Pseudocode

```
START

INPUT P, R, T

PRINT "1. Simple Interest"

PRINT "2. Compound Amount"

PRINT "3. Compound Interest"

PRINT "4. Total Payable Amount"

INPUT choice

SI = (P * R * T) / 100

IF choice = 1 THEN

   PRINT SI

ELSE IF choice = 2 THEN

   A = P * (1 + R/100)^T

   PRINT A

ELSE IF choice = 3 THEN

   A = P * (1 + R/100)^T

   CI = A - P

   PRINT CI

ELSE IF choice = 4 THEN

   Total = P + SI

   PRINT Total

ELSE

   PRINT "Invalid choice"

END IF

STOP
```

4. Flowchart

```
               START
                 ↓
        Input P, R and T
                 ↓
     Display Menu
     1. Simple Interest
     2. Compound Amount
     3. Compound Interest
     4. Total Payable Amount
                ↓
          Input Choice
                ↓
          SI = P*R*T/100
                 ↓
        ◇ Choice = 1? ◇
       /               \
     Yes                No
      ↓                  ↓
   Display SI       ◇ Choice = 2? ◇
                    /              \
                  Yes               No
                   ↓                 ↓
          A=P(1+R/100)^T   ◇ Choice=3? ◇
                   ↓         /           \
             Display A     Yes            No
                            ↓              ↓
                  A=P(1+R/100)^T   ◇Choice=4?◇
                  CI=A-P            /        \
                            ↓     Yes         No
                     Display CI    ↓           ↓
                             Total=P+SI   Invalid
                                   ↓      Choice
                             Display Total
                                   ↓
                                 STOP
```

5. Test Case Table

Test Case	Input(s)	Choice	Expected Output(s)	Actual Output(s)	Result
1	P=1000, R=10, T=2	1	SI = 200.00	SI = 200.00	Pass
2	P=1000, R=10, T=2	2	Amount = 1210.00	Amount = 1210.00	Pass
3	P=1000, R=10, T=2	3	CI = 210.00	CI = 210.00	Pass
4	P=1000, R=10, T=2	4	Total = 1200.00	Total = 1200.00	Pass
5	P=5000, R=5, T=2	3	CI = 512.50	CI = 512.50	Pass

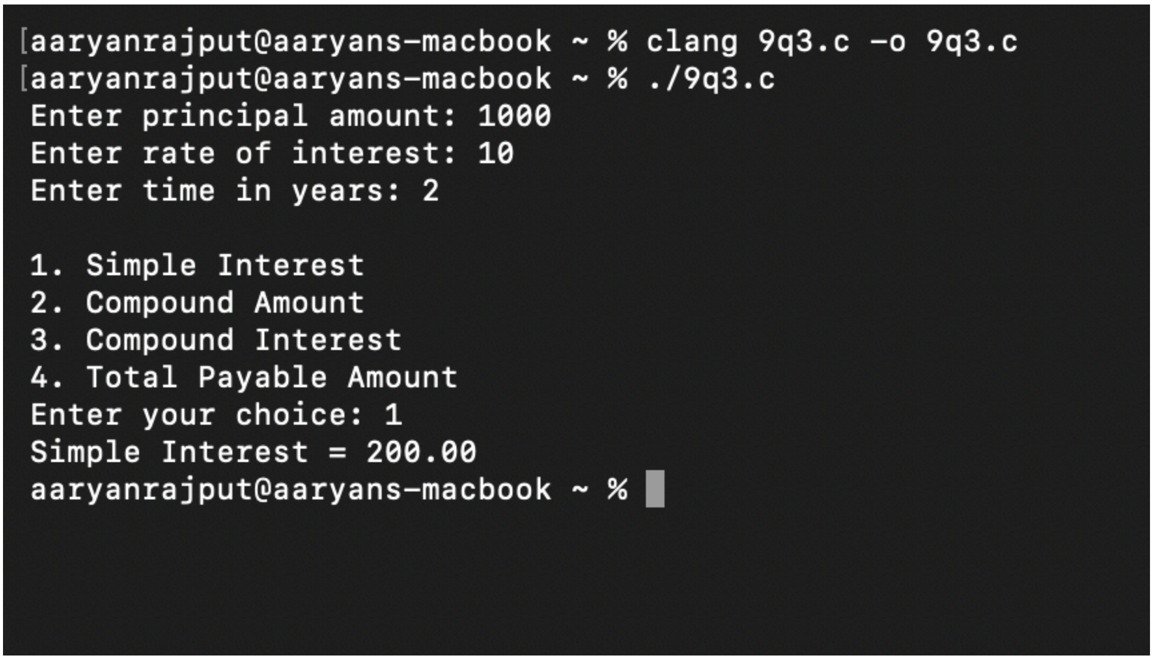
6. C Program

```
#include <stdio.h>

#include <math.h>

int main()
{
    double p, r, t;
    double si, amount, ci, total;
    int choice;
    printf("Enter principal amount: ");
    scanf("%lf", &p);
    printf("Enter rate of interest: ");
    scanf("%lf", &r);
    printf("Enter time in years: ");
    scanf("%lf", &t);
    printf("\n1. Simple Interest");
    printf("\n2. Compound Amount");
    printf("\n3. Compound Interest");
    printf("\n4. Total Payable Amount");
    printf("\nEnter your choice: ");
    scanf("%d", &choice);
    si = (p * r * t) / 100.0;
    switch(choice)
    {
        case 1:
            printf("Simple Interest = %.2lf\n", si);
            break;
        case 2:
            amount = p * pow(1 + r / 100.0, t);
            printf("Compound Amount = %.2lf\n", amount);
            break;
        case 3:
            amount = p * pow(1 + r / 100.0, t);
            ci = amount - p;
            printf("Compound Interest = %.2lf\n", ci);
            break;
        case 4:
            total = p + si;
            printf("Total Payable Amount = %.2lf\n", total);
            break;
        default:
            printf("Invalid choice\n");
    }
```

```
    return 0;
}
```

```
UW PICO 5.09                         File: 9q3.c                         Modified

#include <stdio.h>
#include <math.h>

int main()
{
    double p, r, t;
    double si, amount, ci, total;
    int choice;

    printf("Enter principal amount: ");
    scanf("%lf", &p);

    printf("Enter rate of interest: ");
    scanf("%lf", &r);

    printf("Enter time in years: ");
    scanf("%lf", &t);

    printf("\n1. Simple Interest");
    printf("\n2. Compound Amount");
    printf("\n3. Compound Interest");
    printf("\n4. Total Payable Amount");
    printf("\nEnter your choice: ");
    scanf("%d", &choice);

    si = (p * r * t) / 100.0;

    switch(choice)
    {
        case 1:
            printf("Simple Interest = %.2lf\n", si);
            break;

        case 2:
            amount = p * pow(1 + r / 100.0, t);
            printf("Compound Amount = %.2lf\n", amount);
            break;

        case 3:
            amount = p * pow(1 + r / 100.0, t);
            ci = amount - p;
            printf("Compound Interest = %.2lf\n", ci);
            break;

        case 4:
            total = p + si;
            printf("Total Payable Amount = %.2lf\n", total);
            break;

        default:
            printf("Invalid choice\n");
    }

    return 0;
}

^G Get Help    ^O WriteOut    ^R Read File    ^Y Prev Pg    ^K Cut Text      ^C Cur Pos
^X Exit        ^J Justify     ^W Where is     ^V Next Pg    ^U UnCut Text    ^T To Spell
```

7. Compilation and Execution



```
[aaryanrajput@aaryans-macbook ~ % clang 9q3.c -o 9q3.c
[aaryanrajput@aaryans-macbook ~ % ./9q3.c
Enter principal amount: 1000
Enter rate of interest: 10
Enter time in years: 2

1. Simple Interest
2. Compound Amount
3. Compound Interest
4. Total Payable Amount
Enter your choice: 1
Simple Interest = 200.00
aaryanrajput@aaryans-macbook ~ %
```

Compilation and execution successful.